

Sorbead® Air CCS

Sorbead® Air CCS is a portfolio of high-performance, acid-resistant aluminosilicate gel adsorbents ideally suited for the dehydration of CO₂ for carbon capture and storage.

Product Description

The BASF Sorbead® line of highly efficient adsorbents are aluminosilicate gels in the form of hard, spherical beads, with a very high resistance to crushing and a low attrition rate. Sorbead® adsorbents have a long life, therefore reducing change out costs over time compared to other adsorbents.

Application

Sorbead® Air CCS products are used in temperature swing adsorption (TSA) process to remove impurities from a gas stream. Due to the acid resistance of Sorbead aluminosilicate gels, Sorbead Air CCS products are the ideal solution for the dehydration of acid gases, including CO₂ from point sources, including power plants, refineries, and natural gas plants.

Sorbead adsorbents have higher water capacity at high relative humidity compared to molecular sieves. This high capacity, in combination with BASF proprietary design software and experienced technology managers, lead to smaller TSA units, reducing CapEx. Sorbead adsorbents also require lower regeneration temperatures and are known for their long lifetime between change outs reducing OpEx costs.

Target Properties*

Chemical Composition	96-97% SiO ₂ 3-4% Al ₂ O ₃
Surface Area (BET)	650-700 m ² /g
Pore Volume	0.4 cm ³ /g
Equilibrium Capacity for Water Vapor at 25° C	4-6 wt% at 10 % R.H. 36% at 80 % R.H.
Packed Bulk Density	0.6-0.8 kg/L

*These indicative properties do not represent process capabilities nor specifications

Packaging

- Big Bag: 750-850 kg
- Steel Drum: 130-150 kg

About Us

BASF is a leading global manufacturer of catalysts for the chemical industry, with solutions across the chemical value chain. The business comprises chemical catalysts, adsorbents and custom catalysts. In the process catalyst business, priority is given to developing new and improved products that enable the chemical industry transformation to net-zero emissions.

BASF's chemical catalysts and adsorbent business is part of the company's Performance Chemicals division. The division's portfolio also includes refinery catalysts, fuel and lubricant solutions, as well as oilfield chemicals and mining solutions. Customers from a variety of industries including Chemicals, Plastics, Consumer Goods, Energy & Resources and Automotive & Transportation benefit from our innovative solutions.

BASF - We create chemistry

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